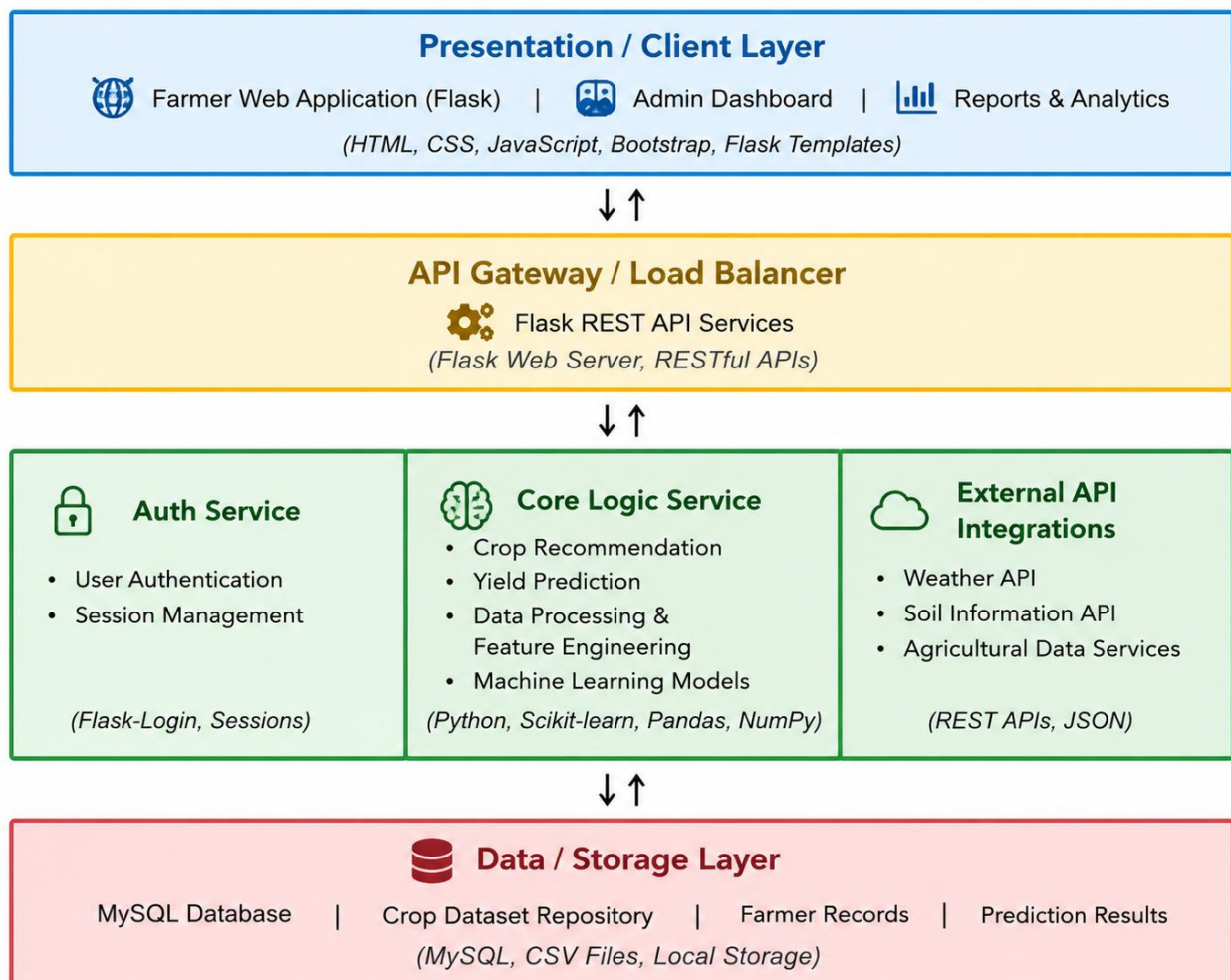


Solution Architecture

Date	04 July 2026
Team ID	XXXXXX
Project Name	Opticode-Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides user interface for farmers and administrators to enter agricultural data and view recommendations.</i>	<i>HTML, CSS, JavaScript, Flask</i>
<i>API Gateway</i>	<i>Handles communication between frontend and backend services through REST APIs.</i>	<i>Flask REST API</i>
<i>Core Logic Service</i>	<i>Processes input data, performs crop recommendation and yield prediction using machine learning models.</i>	<i>Python, Scikit-learn, Pandas, NumPy</i>
<i>External API Integration</i>	<i>Fetches weather and agricultural information to improve prediction accuracy.</i>	<i>Weather API, Agricultural APIs</i>